
ECON 312 — Intermediate Microeconomics

Lecture Notes — Ch. 2: The Budget Constraint

Fall 2026

The purpose of this first part of the course is to understand how consumers make decisions. Before diving into what consumers *do* choose, let's find what they *can* do: these are the things inside their “budget constraint” → what they can afford to do.

Outline

- ① How to draw a budget constraint
- ② How price and budget changes affect the constraint
- ③ How unusual interventions change the constraint

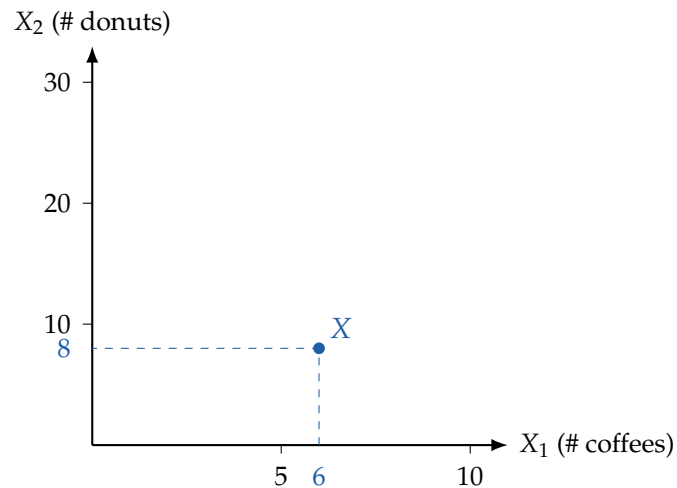
1. How to Draw a Budget Constraint

We'll start in a 2-good market:

- Consumer has budget M (← for *Money*), i.e. \$50
 - ↪ how much \$\$\$ they have to spend
- Price of good 1 is P_1 , good 2: P_2
 - ↪ i.e. \$5/coffee, \$2/donut
- Consumer is choosing a “consumption bundle”
 - ↪ X_1 of good 1, i.e. 6 coffees
 - ↪ X_2 of good 2, i.e. 8 donuts

denoted (X_1, X_2) , or just X depending on context

We can visualize a bundle on an x - y graph:



Is Bundle X “Affordable”?

Q Could our consumer buy it?

Use $P \cdot Q$ (just for each good).

$$6 \text{ coffees} \times \$5/\text{coffee} + 8 \text{ donuts} \times \$2/\text{donut} = \$30 + \$16 = \$46 \checkmark$$

Yes, they can!

→ Note: we didn’t ask “would” or “should”

Q But this gets tedious. Is there an easy way to describe all such points?

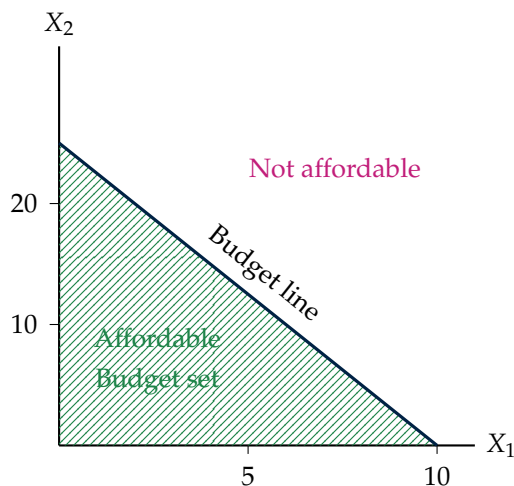
Use an inequality!

$$\underbrace{\overbrace{P_1 X_1}^{\text{expenditure on good 1}} + \overbrace{P_2 X_2}^{\text{expenditure on good 2}}}_{\text{total expenditure}} \leq \underbrace{M}_{\text{budget}} \quad \uparrow \text{ just uses same machinery}$$

less than

Graphing the Full Budget Set

Let’s do one better. Let’s graph the *budget set*:



Properties:

→ Rewrite the budget inequality:

$$\left. \begin{aligned} P_1 X_1 + P_2 X_2 &\leq M \\ \Rightarrow P_2 X_2 &\leq M - P_1 X_1 \\ \Rightarrow X_2 &\leq \frac{M}{P_2} - \frac{P_1}{P_2} X_1 \end{aligned} \right\} \begin{array}{l} \text{I don't memorize this.} \\ \text{Just learn the intuition.} \end{array}$$

Slope is $-\frac{P_1}{P_2}$
 $\Rightarrow$ this is the opportunity cost

So the X_2 -intercept is $\frac{M}{P_2}$, the X_1 -intercept is $\frac{M}{P_1}$.

→ Pretend you “spend all money on X_2 or X_1 ”

Beyond Two Goods

But wait — this is slick, but it only works with 2 goods. Real humans consume more than just coffee & donuts!

Q Is this problematic?

→ No. Let X_1 be the good of choice, and then let X_2 be “everything else,” i.e. just dollars.

The price of “everything else” is just “\$1,” because it is just money for everything else. So here $P_2 = 1$.

↪ when $P_n = 1$, we call n the **numeraire**

(but you could do more than two goods if you want... it just takes more math)

Q Do these prices have to be in dollars?

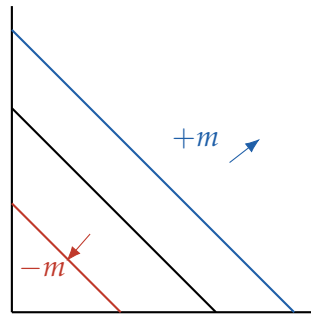
→ No! It can be time, energy, computational time, food, etc...

2. How Price and Budget Changes Affect the Constraint

Use handout

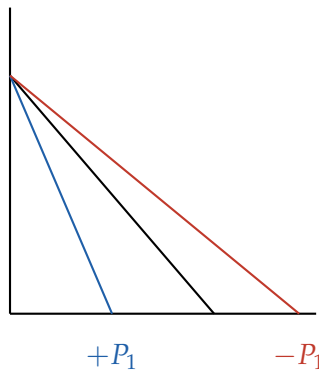
$$\text{Budget Constraint: } P_1 X_1 + P_2 X_2 \leq M$$

Q What if M changes? How does this change the picture?



Changing M shifts the whole BC out or in!

Q What if P_1 changes? Changing P_1 changes the slope.



3. Unusual Interventions

Now we turn to several different ways in which we may get non-linear budget constraints.

→ Purposes:

1. get experience with budget sets
2. get comfortable with several real-world complexities

While going through the hand-out, keep asking:

Q Is the consumer better off? How do we know?